GLUING MAPS OF SHEAVES OF SETS FROM COMPATIBLE LOCAL MAPS

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Allowed standard prerequisites (editorial).

- Sheaf axioms on X: sections agreeing on the members of an open covering glue uniquely. invoked at proof lines 5326-5338. Provides existence and uniqueness of the section phi_U(s) from the compatible local images.

Lemma 0.2. *Let X be a topological space. Let $X = \bigcup U_i$ be an open covering. Let $\mathcal{F}$, $\mathcal{G}$ be sheaves of sets on X. Given a collection*

$$\varphi_i : \mathcal{F}|_{U_i} \longrightarrow \mathcal{G}|_{U_i}$$

of maps of sheaves such that for all $i, j \in I$ the maps φ_i, φ_j restrict to the same map $\mathcal{F}|_{U_i \cap U_j} \to \mathcal{G}|_{U_i \cap U_j}$ then there exists a unique map of sheaves

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

whose restriction to each U_i agrees with φ_i.

Proof. For each open subset $U \subset X$ define

$$\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U), \quad s \mapsto \varphi_U(s)$$

where $\varphi_U(s)$ is the unique section verifying

$$(\varphi_U(s))|_{U \cap U_i} = (\varphi_i)_{U \cap U_i}(s|_{U \cap U_i}).$$

Existence and uniqueness of such a section follows from the sheaf axioms due to the fact that

$$\begin{aligned} ((\varphi_i)_{U \cap U_i}(s|_{U \cap U_i}))|_{U \cap U_i \cap U_j} &= (\varphi_i)_{U \cap U_i \cap U_j}(s|_{U \cap U_i \cap U_j}) \\ &= (\varphi_j)_{U \cap U_i \cap U_j}(s|_{U \cap U_i \cap U_j}) \\ &= ((\varphi_j)_{U \cap U_j}(s|_{U \cap U_j}))|_{U \cap U_i \cap U_j}. \end{aligned}$$

This family of maps gives us indeed a map of sheaves: Let $V \subset U \subset X$ be open subsets then

$$(\varphi_U(s))|_V = \varphi_V(s|_V)$$

Date: stacks-project:sheaves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

since for each $i \in I$ the following holds

$$\begin{aligned}(\varphi_U(s))|_{V\cap U_i} &= ((\varphi_U(s))|_{U\cap U_i})|_{V\cap U_i} \\ &= ((\varphi_i)_{U\cap U_i}(s|_{U\cap U_i}))|_{V\cap U_i} \\ &= (\varphi_i)_{V\cap U_i}(s|_{V\cap U_i}) \\ &= \varphi_V(s_V)|_{V\cap U_i}.\end{aligned}$$

Furthermore, its restriction to each U_i agrees with φ_i since given $U \subset X$ open subset and $s \in \mathcal{F}(U \cap U_i)$ then

$$\begin{aligned}\varphi_{U\cap U_i}(s) &= \varphi_{U\cap U_i}(s)|_{U\cap U_i} \\ &= (\varphi_i)_{U\cap U_i}(s|_{U\cap U_i}) \\ &= (\varphi_i)_{U\cap U_i}(s).\end{aligned}$$

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